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INTRODUCTION

Senior Robotics Engineer with 7+ years of experience in **aerial, mobile, and multi-robot systems**, spanning simulation, autonomy, and real-world deployments. Currently working toward **safe multi-drone coordination, collision avoidance, and persistent area coverage** for real-world applications. Driven to build production-ready autonomy and motivated to **bridge the gap between multi-robot systems research and deployable robotic systems**, particularly in safety-critical domains.

TECHNICAL SKILLS

Languages: C++, Python

Softwares: ROS, Linux, Git, Gazebo, Docker, Rviz, Latex, Doxygen, MoveIt, ATAK

Courses & Concepts: Swarm Intelligence, Localization, Path Planning, Perception, Algorithms, Mobile Robotics, Aerial Robotics, Robot Controls & Dynamics, OOP, SLAM, Behavior Trees, Autonomy, Obstacle Avoidance, Navigation, Data Structures, Sensor Fusion, State Machines

EDUCATION

Worcester Polytechnic Institute (WPI)

Master of Science in Robotics Engineering

GPA: 4.0/4.0

Worcester, MA

May 2019

University of Pune (UoP)

Bachelor of Mechanical Engineering

Percentage: 71/100

Pune, India

June 2016

WORK EXPERIENCE

Nightingale Security *Senior Robotics Software Engineer*

March 2024 – Present

- Technical owner for **ATAK integration** with Nightingale Drones, implementing and deploying **multi-drone control and monitoring from a single ATAK client**, and maintaining compatibility with latest ATAK version.
- Developing **multi-drone aerial robotics systems designed** for continuous security operations, with emphasis on system reliability, robustness, and operational safety.
- Architected and deployed production-grade **UAV simulation infrastructure**, enabling safe, large-scale testing, development, and system validation without reliance on physical hardware.
- Led **end-to-end drone provisioning, automation, and deployment readiness**, including testing, configuration, and documentation, eliminating single-point dependencies and standardizing operational workflows.
- Delivering sustainable high-priority **customer support** for **defense, law-enforcement**, and energy partners, including direct technical coordination, **training**, and support for **24/7 security operations**.
- **Designed** and implemented a structured **customer-support** and **on-call model**, assigning clear ownership of customers to team members, improving response efficiency, ensuring fairness in workload, and enabling the team to grow by developing independent expertise across all customer accounts.
- Enabled team scalability through **technical mentoring, documentation, and knowledge transfer**, serving as an escalation point for complex system-level issues
- Contributed to **hiring** and **team growth** through resume screening, interview planning, and technical evaluations.

ClearPath Inc *Senior Simulation Developer-Robotics*

Feb 2023 – July 2023

- Developed simulations of robots in **Gazebo** using **AWS EC2 instances** for real-world applications.
- Imported robot **STL models** into gazebo simulation, enabling dynamic parameter changes to modify robot functionality.
- Led the planning and implementation of a new **pallet pick-and-place feature** for a project.
- Delivered presentations to executives, effectively communicating project progress and feature implementation details.
- Acted as a point of contact and triage person for the team, addressing issues from other teams and stakeholders promptly and creating tickets for further team attention.
- Collaborated with cross-functional teams to troubleshoot and resolve technical challenges, ensuring seamless project execution and delivery.

Open Robotics *Software Engineer-Robotics*

Aug 2022 – Dec, 2022

- Successfully verified and executed build instructions and tutorials for **ROS/Gazebo** (garden release), demonstrating a deep understanding of Gazebo package architecture.

- Implemented bug fixes in **Gazebo rendering packages** regarding **lens flare** from **wide-angle camera**, as a part of the **NASA VIPER team**.
- Conducted thorough code reviews and comprehensive testing of contributions from individual open-source contributors to **gazebo-sim**, ensuring code quality and reliability.

Symbotic LLC *Advanced Controls Engineer*

Aug 2019 – Aug 2022

- Spearheaded projects for automated mobile robots, specializing in **case under pick ability**.
- Developed advanced tools for real-time issue diagnosis and troubleshooting by **analyzing sensor data** from **AMR**.
- Implemented and tested improvements to existing features, considering future scalability and expansion of robot capabilities.
- Optimized code scalability to handle **simulations** with over **100 robots simultaneously**, addressing robot peculiarities.
- Enhanced robot performance by identifying and resolving limitations in specific scenarios.
- Collaborated with multiple teams to diagnose and solve **localization** problems, ensuring reliable results.
- Contributed to the development and deployment of pick/place algorithms, utilizing both conventional and **AI-based control strategies**.
- Collaborated with the test team in a test-driven environment to validate algorithms.
- Programmed, unit tested, debugged, and improved pick/place ability through troubleshooting and optimization.

Waypoint Robotics Inc *Robotics Intern*

Aug, 2018 – Dec, 2018

- Assembled an Autonomous Mobile Robot (AMR) and programmed its behavior utilizing **LIDAR data** and digital IO for bystander feedback regarding robot intentions.
- Enhanced the robot's programming environment by implementing **sensor fusion** and motion planning, introducing novel features to the product's graphical user interface (GUI).
- Integrated a **detection deep learning algorithm** on live video input from a mobile robot's camera, pioneering a motion algorithm that responds to this input effectively.

Cere Labs Pvt Ltd *Machine Learning Intern*

March, 2016 – June, 2016

- **Applied Reinforcement Learning (RL)** techniques, particularly the **Q-learning algorithm**, to autonomously guide a crawling robot toward a wall using ultrasonic sensor data and **Raspberry Pi** controller.
- Documented the comprehensive process of constructing the robot and executing the reinforcement learning algorithm, showcasing expertise in technical communication, and published a series of informative blog posts on the company's platform.

PROJECTS (URL)

Collective transport of Concave objects using a robot swarm *Swarm Intelligence* Jan – May, 2018

- Devised and executed an innovative occlusion-based collective transport strategy using **Khepera IV** robots in **ARGoS**, transforming concave objects to convex by strategic filling in C++.
- Achieved **efficient object transport without prior knowledge** or explicit inter-robot communication, highlighting the strategy's autonomy and adaptability.

Automated Cinematography using an UAV *Motion Planning*

Aug – Dec, 2018

- Employed a **hector quadrotor** to execute a comprehensive **360-degree aerial cinematography** view of a human subject within a simulated environment.
- Implemented a **ROS-based local RRT* path planner** on a quadrotor, ensuring **obstacle avoidance**, and captured environment images using **RViz** for precise path validation.

Atlas's Escape *Humanoid Robotics*

Jan – May, 2019

- Developed Atlas humanoid robot's behavior by integrating **perception**, **locomotion**, and **manipulation** to autonomously detect and approach doors in Gazebo using ROS and C++.
- Utilized **BGR images** and **LiDAR point cloud data** for accurate door sensing and recognition.

Detection, Recognition, Pose Estimation of Objects *Deep Learning for Perception*

Aug–Dec, 2017

- Achieved **98% accuracy** in **object recognition** and an average of **85% in angle estimation** on the TableTop dataset.
- Successfully processed images from **8 distinct angles** and **2 varied heights** for comprehensive results.

EXTERNAL IMPACT

Co-Organizer Women in Robotics Boston Chapter

March, 2020 – June, 2022

Organizer Women in Robotics Bay Area

July, 2023 – Present

Organizing events ranging from interviews of inspiring women working in the field to technical events digging to the core of various robotics concepts.

FIRST Robotics

March, 2025 – Present

Technical judge at local FIRST robotics competitions.